

Description of Work- Trail Maintenance Scenario

Trail history and condition:

McKenzie Gulch 9W85, TC3: was constructed 10 years ago, and was burned over by the Oregon Fire in 2018. This trail is managed for bicycle use and is open to hiker and equestrian users. The trail tread is damaged the entire length of the trail. Berms and other bike features will need to be repaired. Heavy brushing and hazard tree removals need on approximately 1 mi of the trail.

McKenzie Gulch Spur 9W86, TC3: is a user created trail brought into the system. It needs tread maintenance and brushing. This section lacks the needed erosion control features needed to keep water from running down the trail. There are hazard trees and logs across the trail that need to be removed. This trail is designed for hiker and equestrian users. Bikes are not allowed. Mechanized equipment can be used on the first 2 miles of trail, but hand work is required on the bottom .1 miles. Equipment is not allowed on the bottom 0.1 miles, so equipment will have to be tracked out.

Trail Design Parameters (from TMO):

McKinzie Gulch 9W85 2.88 Miles

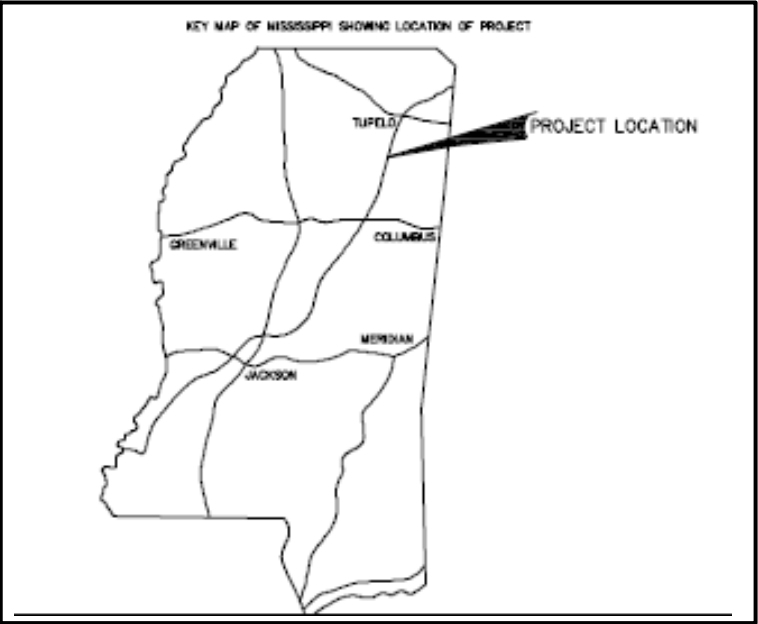
18" wide trail tread
10' clearing limit height
72" clearing limit width
Typical Grade 10%
Maximum Short Pitch 15%
Non-wilderness

McKinzie Gulch Spur 9W86 2.10 Miles

18" wide trail tread
10' clearing limit height
72" clearing limit width
Typical Grade 8%
Maximum Short Pitch 15%
Non-wilderness

Methods:

Mechanized construction allowed, except where noted above. Area will be flagged for avoidance. Noise restrictions from Feb 1st to July 30th for 2 hours per day (chain saw only). Soil has a high clay component and construction will be best implemented May-June or October-November.



UNITED STATES DEPARTMENT OF AGRICULTURE



FOREST SERVICE



SMOKEY BEAR NATIONAL FOREST

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TRAIL CONSTRUCTION PLANS FOR

TRAIL
MAINTENANCE SCENARIO

INDEX

SHEET

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REVIEWED BY:

XXX DATE

XXX DATE

RECOMMENDED BY:

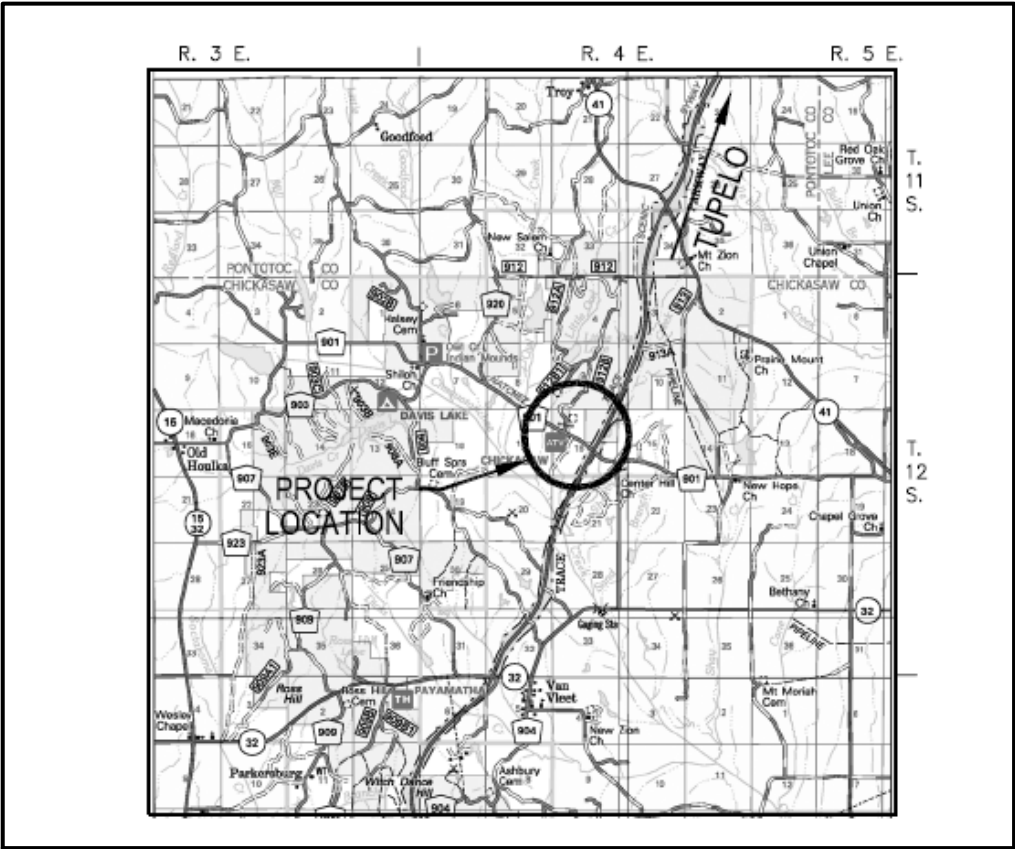
DISTRICT RANGER/PROGRAM MANAGER DATE

FOREST/STATION ENGINEER DATE

APPROVED BY:

FOREST SUPERVISOR/STATION DIRECTOR DATE

REGIONAL/RESEARCH ENGINEER DATE



PROJECT SITE

REVISIONS:

No. 1	_____	DATE	INITIALS
No. 2	_____	DATE	INITIALS
No. 3	_____	DATE	INITIALS

DESIGN:	_____	DATE
DRAWN:	_____	DATE
CHECKED:	_____	DATE

PROJECT NAME

SAMPLE TRAIL MAINTENANCE

SECTION

900-GENERAL

TYPICAL ID

TTL

REVISION DATE

XX/XX/XX

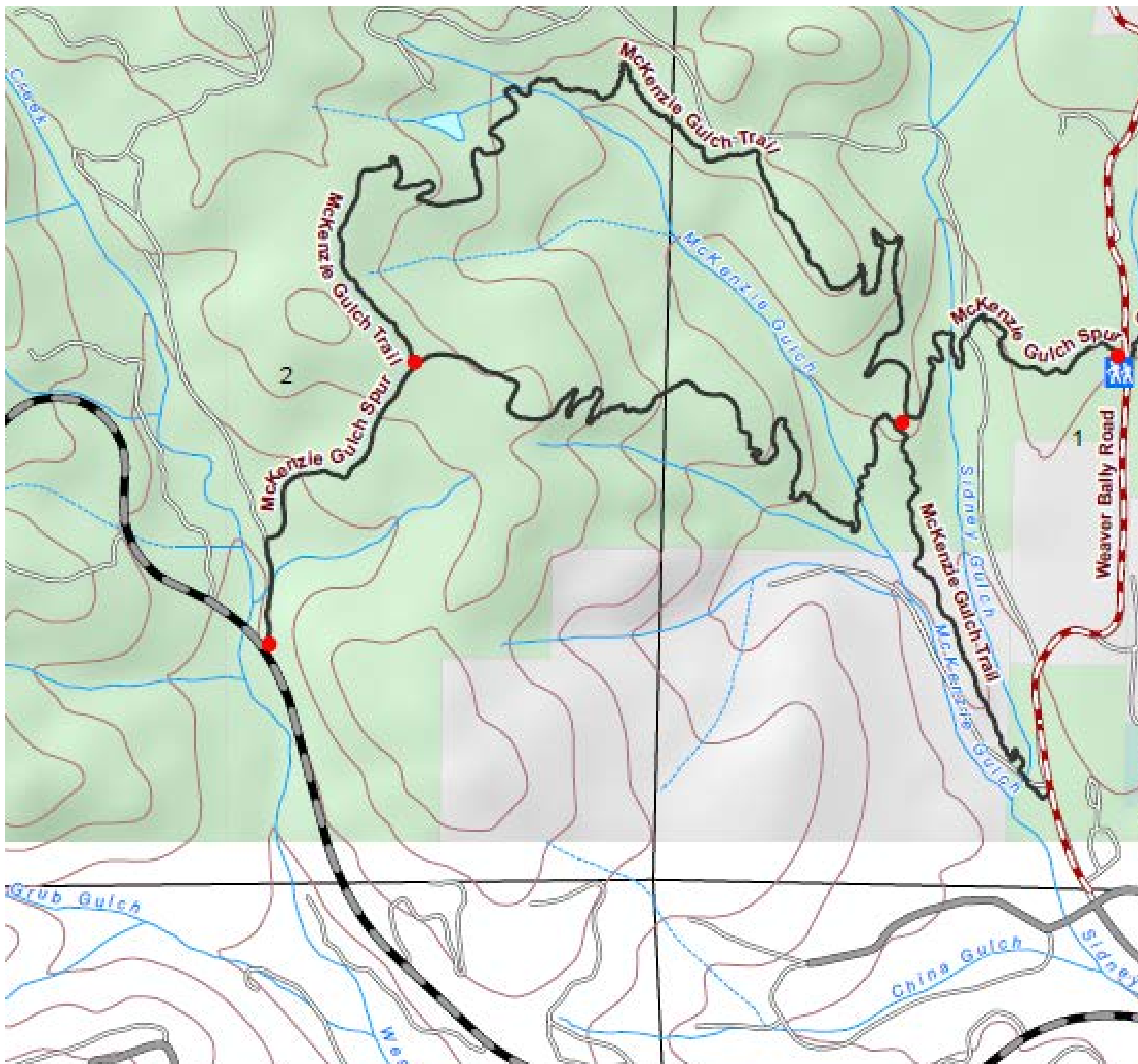
NO SCALE

DRAWING NO.

STD_900-01

SHEET

OF



TYPICAL ID DEFINITIONS

GENERAL NOTES

DRAWING NUMBER	TYP ID	DRAWING NAME	DRAWING NUMBER	TYP ID	DRAWING NAME
STD_900-01	TTL	TITLE_SHEET	STD_932-10-01	TPK	TURNPIKE
STD_900-02	VCM	VICINITY_MAP	STD_932-20-01	TPF	TURNPIKE_W/FOUNDATION
STD_900-03	GNT	GENERAL_NOTES	STD_933-10-01	SRB	STACKED_ROCK_BARRIER
STD_900-04	WKL	WORK_LIST	STD_933-20-01	MRB	MASONRY_ROCK_BARRIER
STD_900-05	LD1	LINE_DIAGRAM	STD-933-20-02	CMB	CONCRETE_MASONRY_UNIT_BARRIER
STD_900-06	LD2	LINE_DIAGRAM	STD_933-30-01	BRG	BARRIER_RAIL_ON_GRADE
STD_900-07	PP1	PLAN_PROFILE_10	STD_933-40-01	BRP	BARRIER_RAIL_ON_POST
STD_900-08	PP2	PLAN_PROFILE_20	STD_933-40-02	TBP	TIMBER BARRIER ON POST
STD_906-01	SOQ	SUMMARY_OF_QUANTITIES	STD_933-50-01	CRB	CURBS
STD_910-01	STT	STANDARD_TRAIL_TERMS	STD_933-60-01	WGR	GUARDRAIL
STD_911-01	TSF/TCS	TYPICAL_CROSS_SECTION_SLOPE_FINISH	STD_934-10-01	SP1	STANDARD_PUNCHEON
STD_911-02	TDD/TOC	TYPICAL_OBSTRUCTION_CLEARANCE_SECTION	STD_934-10-02	SP2	STANDARD_PUNCHEON
STD_911-03	TRT	TYPICAL_RETAINERS	STD_934-20-01	NDP	NO_DECK_PUNCHEON
STD_911-30-01	ETR	EXISTING_TRAIL_RESTORATION	STD_935-10-01	LCB	LOG_CRIB
STD_911-60-01	OBT	OBLITERATION_OF_TRAILS	STD_935-20-01	RRW	STACKED_ROCK_RETAINING_WALL
STD_912-01	CLT	CLEARING_LIMITS_TREES_LOGS	STD_935-30-01	BRW	GABION_BASKET_RETAINING_WALL
STD_912-02	CLB	CLEARING_LIMITS_BRUSHING	STD_935-40-01	MRW	MASONRY_ROCK_RETAINING_WALL
STD_913-01	TSS	TYPICAL_SURFACING_SECTION	STD_935-40-02	CUM	CONCRETE_MASONRY_UNIT_RETAINING_WALL
STD_913-02	TSR	TYPICAL_SURFACING_SECTION_W/RETAINERS	STD_935-50-01	CCW	CAST_INPLACE_CONCRETE_WALL
STD_913-03	THS	TYPICAL_HARDENED_SURFACING	STD_936-10-01	ITS	INDIVIDUAL_TIMBER_PINNED_STEP
STD_913-04	HSR	TYPICAL_HARDENED_SURFACING_W/RETAINERS	STD_936-10-02	IRS	INDIVIDUAL_ROCK_STEP
STD_913-40-01	GUS	GRID_UNIT_SURFACING	STD_936-20-01	ORS	OVERLAPPING_ROCK_STEP
STD_913-50-01	RRS	RIPRAP_SURFACING	STD_936-20-02	OTS	OVERLAPPING_TIMBER_STEP
STD_914-01	CTN	CLIMBING_TURN	STD_936-30-01	CRL	CRIB_LADDER
STD_915-01	TAS	TALUS_SECTION	STD_936-40-01	STC	STAIRCASE
STD_916-01	TPS/TOS	TURNOUT_&_PASSING_SECTION	STD_937-10-01	RLS	HANDRAIL_SYSTEM
STD_917-10-01	NFD	NATURAL_FORD	STD_938-10-01	SB1	STANDARD_BOARDWALK
STD_917-20-01	RF1	CONSTRUCTED_ROCK_FORD	STD_938-10-02	SB2	STANDARD_BOARDWALK
STD_917-20-02	RF2	CONSTRUCTED_ROCK_FORD	STD_938-10-03	SB3	STANDARD_BOARDWALK
STD_917-30-01	LF1	CONSTRUCTED_LOG_FORD	STD_938-20-01	EB1	ELEVATED_BOARDWALK
STD_917-30-02	LF2	CONSTRUCTED_LOG_FORD	STD_938-20-02	EB2	ELEVATED_BOARDWALK
STD_918-10-01	RKF	ROCK FOUNDATIONS	STD_938-20-03	EB3	ELEVATED_BOARDWALK
STD_918-20-01	GBF	GABION BASKET FOUNDATIONS	STD_938-20-04	EB4	ELEVATED_BOARDWALK
STD_918-30-01	CBF	CRIB FOUNDATIONS	STD_938-30-01	SNR	STEP_N_RUN
STD_918-40-01	GEF	GEOSYNTHETIC FOUNDATIONS	STD_941-10-01	PWF	POST_N_WIRE_FENCE
STD_918-50-01	CDF	CORDUROY FOUNDATIONS	STD_941-20-01	PRF	POST_N_RAIL_FENCE
STD_921-10-01	SCV	STANDARD_CULVERTS	STD_941-30-01	WWF	WOVEN_WIRE_FENCE
STD_921-20-01	CVH	STANDARD_CULVERTS_W/HEADWALLS	STD_941-40-01	JLF	JACK_LEG_FENCE
STD_921-30-01	RCV	ROCK_CULVERT	STD_941-50-01	SRF	STACKED_RAIL_(WORM)_FENCE
STD_921-40-01	TCV	TREATED_TIMBER_BOX_CULVERT	STD_942-10-01	WRG	WIRE_GATE
STD_921-50-01	OTD	OPEN_TOP_DRAIN	STD_942-10-02	WWG	WOVEN_WIRE_GATE
STD_921-60-01	ACV	BOTTOMLESS_ARCH_ROCK_HEADWALL	STD_942-20-01	MSG	METAL_SWING_GATE
STD_921-70-01	LCV	LOG_CULVERTS	STD_942-20-02	WSG	WOOD_SWING_GATE
STD_922-10-01	RWB	ROCK_WATERBAR	STD_942-20-03	WKG	WOOD_SWING_GATE (FOR KISSING GATES)
STD_922-20-01	TWB	LOG_OR_TREATED_TIMBER_WATERBAR	STD_942-30-01	LRG	LOOSE_RAIL_GATE
STD_922-30-01	BWB	BELTED_WATERBAR	STD_942-40-01	KG1	ACCESSIBLE_KISSING_GATE_IN_FENCE
STD_923-10-01	RSP	ROCK_SPILLWAY	STD_942-40-02	KG2	ACCESSIBLE_KISSING_GATE_ON_FENCE
STD_924-10-01	RUD	ROCK_UNDERDRAIN	STD_942-50-01	CHI	ACCESSIBLE_CHICANES
STD_924-20-01	SUD	SHEET_UNDERDRAIN	STD_944-10-01	STL	STILES
STD_924-20-02	SDP	SHEET_UNDERDRAIN_W/PIPE	STD_945-10-01	BLD	BOLLARDS
STD_925-01	TDS	TYPICAL_DITCH_SECTIONS	STD_950-01	SNM	SIGNS_&_MARKERS
STD_926-01	BRM	BERMS	STD_951-01	LIN	LOCATION_&_INFORMATION
STD_927-10-01	DD1	DRAIN_DIP	STD_951-02	SPI	SIGN_&_POST_INSTALLATION
STD_927-10-02	DD2	DRAIN_DIP_DETAILS	STD_952-01	RTM	ROUTE_MARKERS
STD_928-10-01	CKD	CHECK_DAM	STD_953-01	RSM	REASSURANCE_MARKER
STD_931-10-01	SW1	SWITCHBACK_TYPE_1	STD_955-01	RCC	ROCK_CAIRN_CONSTRUCTION
STD_931-20-01	SW2	SWITCHBACK_TYPE_2	STD_981-01	SMF	SEEDING_MULCHING_&_FERTILIZING
STD_931-30-01	SW3	SWITCHBACK_TYPE_3			

1. DRAIN DIPS AS STAKED IN THE FIELD
2. EQUIPMENT EXCLUSION AREAS FLAGGED WITH YELLOW/BLACK STRIPPED FLAGGING

WORK LIST

[illegible]

PROJECT NAME & LOCATION

DRAWING NAME	WORK LIST	
SECTION	900 - GENERAL	TYPICAL ID WKL

REVISION DATE
XX/XX/XX

NO SCALE

DRAWING NO.	
STD_900-04	
SHEET	OF

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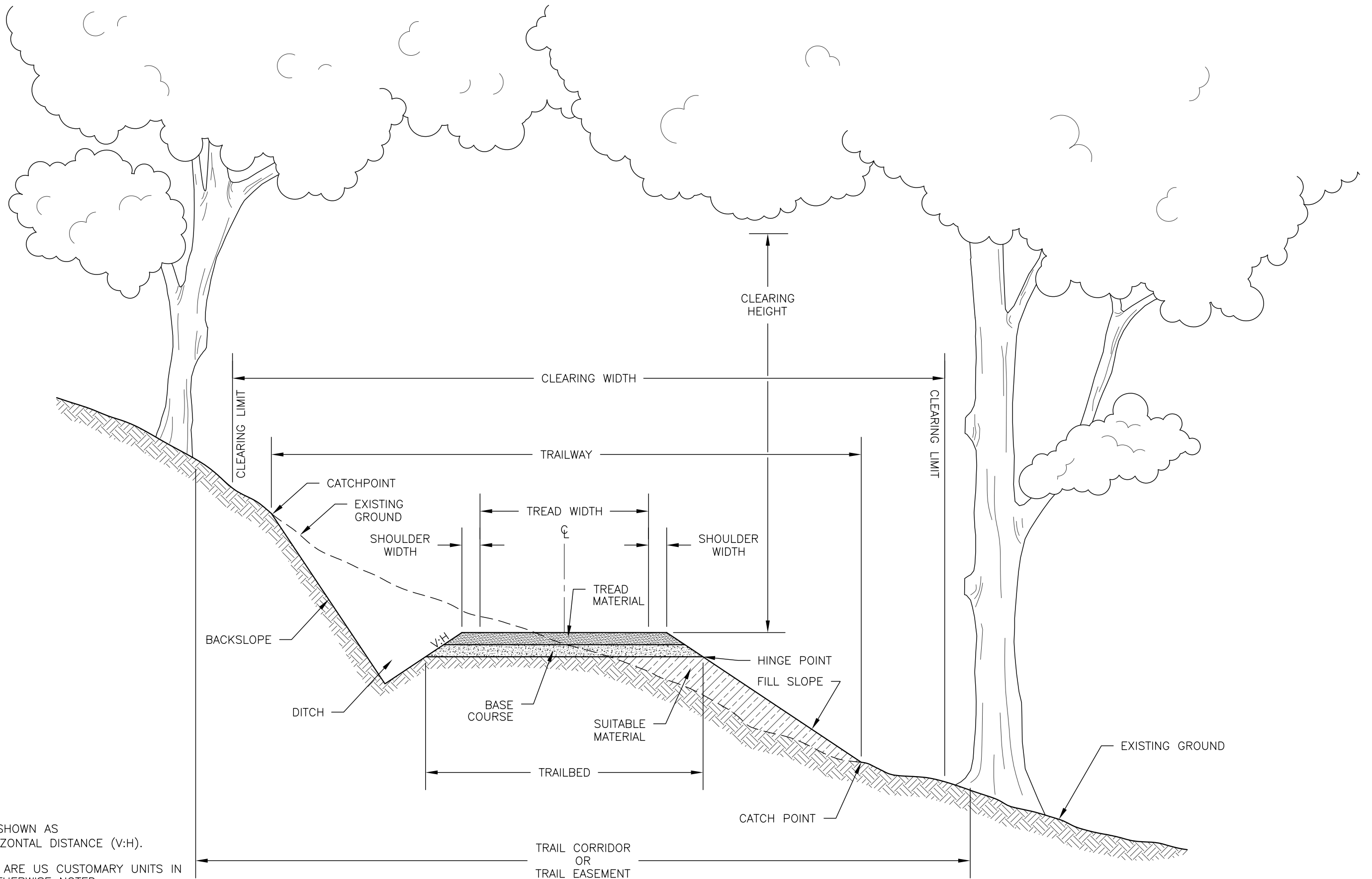
PROJECT NAME & LOCATION	

DRAWING NAME	
SUMMARY OF QUANTITIES	
SECTION	TYPICAL ID
906 - MEASUREMENT & PAYMENT	SOQ

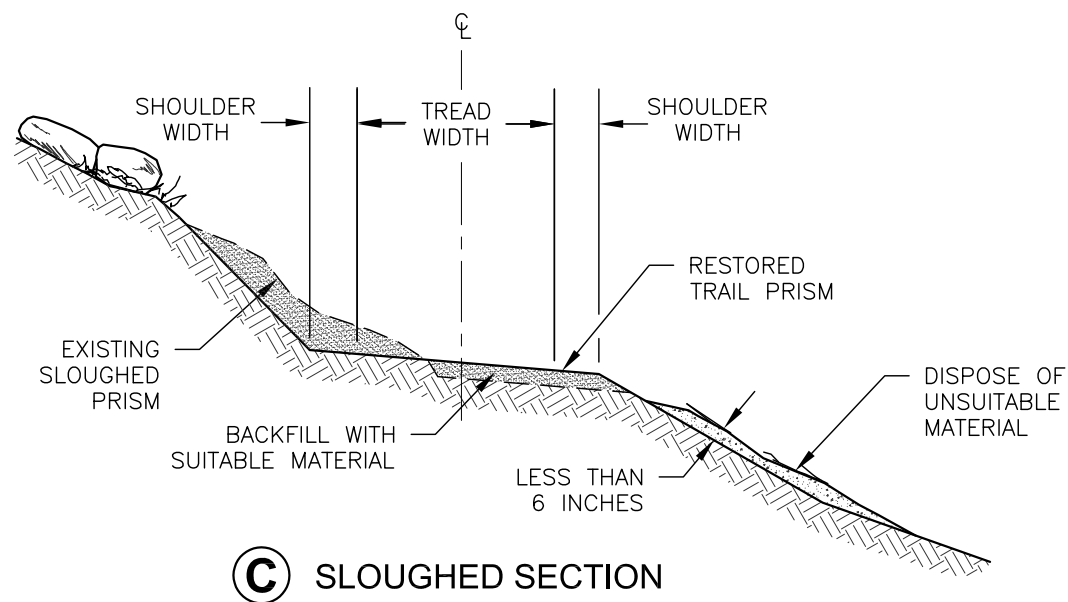
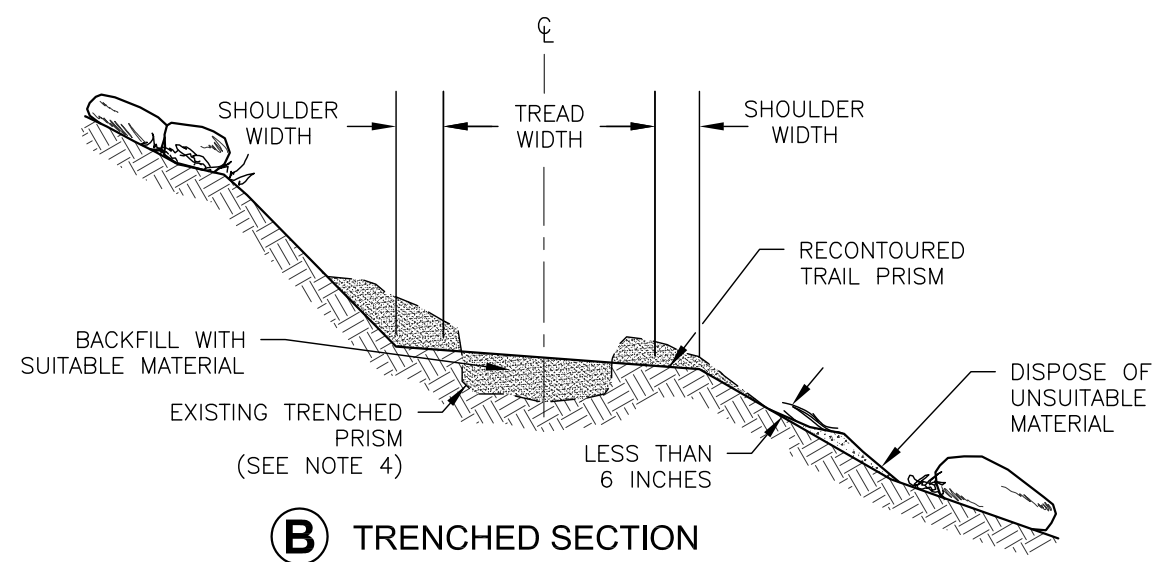
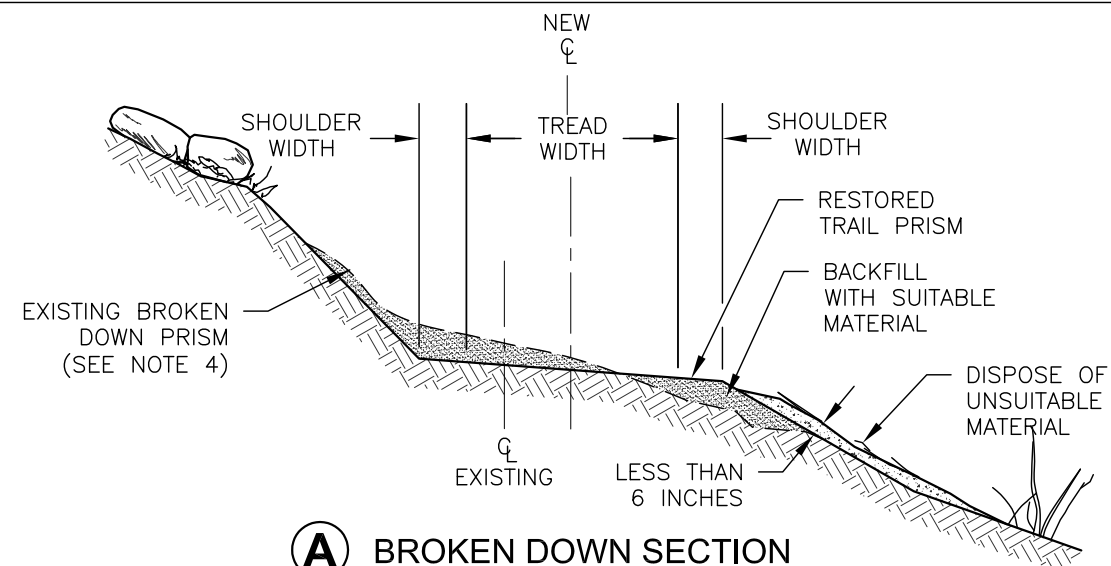
REVISION DATE
XX/XX/XX

NOT TO SCALE

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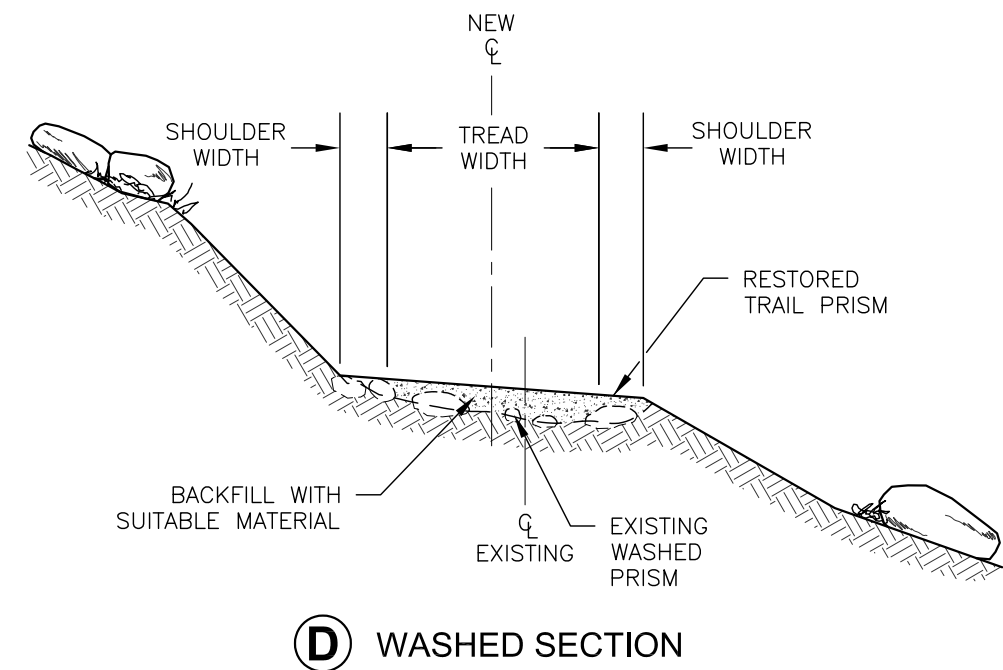
NOTES:
ALL SLOPES ARE SHOWN AS
VERTICAL-TO-HORIZONTAL DISTANCE (V:H).
ALL UNITS SHOWN ARE US CUSTOMARY UNITS IN
INCHES UNLESS OTHERWISE NOTED.



EXISTING TRAIL RESTORATION					
TYPICAL ID	SECTION TYPE	TREAD WIDTH	SHOULDER WIDTH		COMMENTS
			UPHILL	DOWNHILL	
ETR-1	ALL	18 IN	12 IN	12 IN	

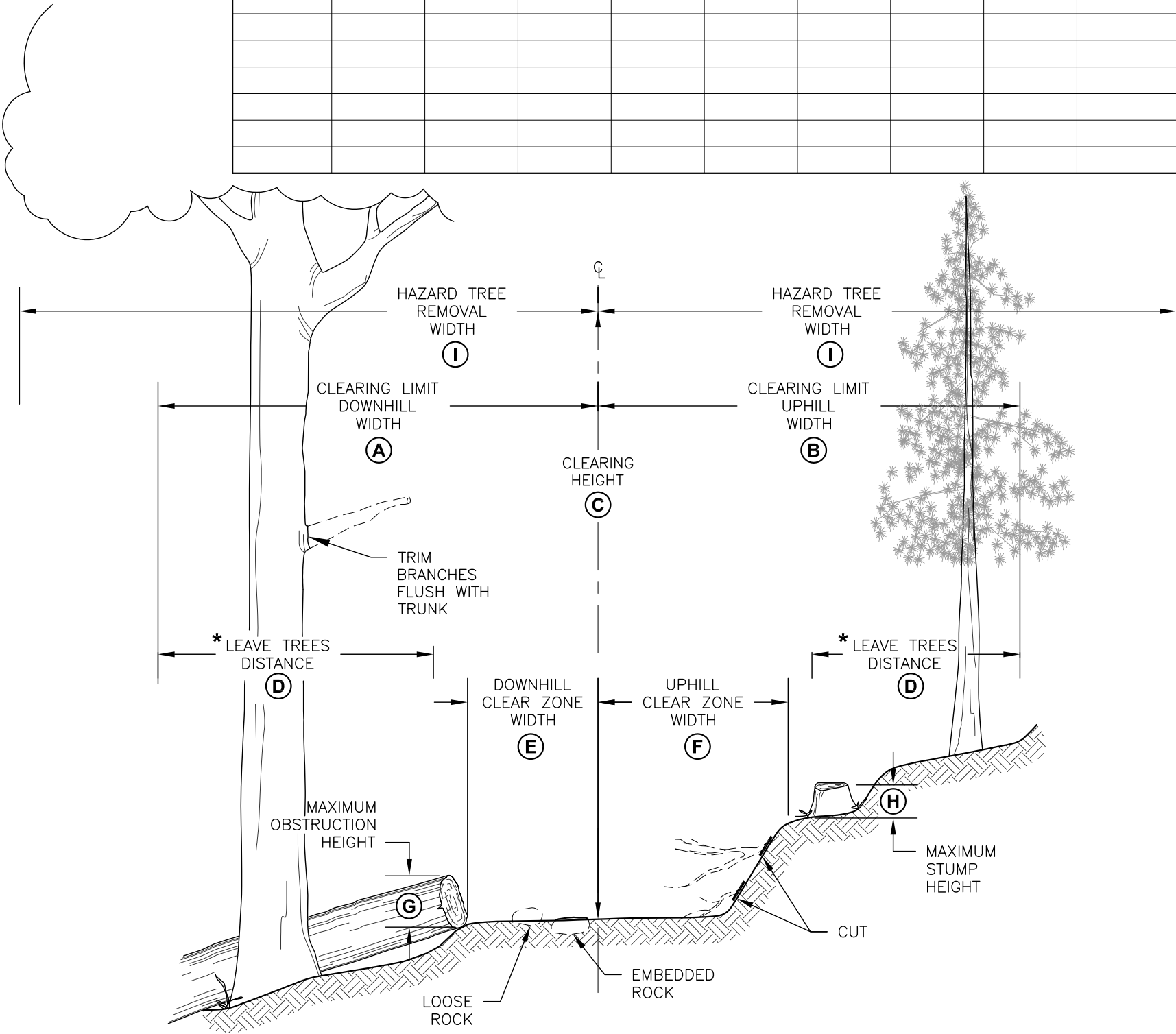
NOTES:

1. RE-ESTABLISH ORIGINAL DRAINAGE STRUCTURES TO MATCH NEW TREAD SURFACE.
2. INSTALL CHECK DAMS, DRAINAGE DIPS OR OTHER DRAINAGE STRUCTURES WHEN SPECIFIED.
3. DRAINAGE DIPS WILL BE STAKED IN THE FIELD WHEN REQUIRED AND WILL BE PAID SEPARATELY UNDER SECTION 927.
4. USE ONLY SUITABLE MATERIAL TO CONSTRUCT RESTORED TRAIL PRISMS. DISPOSE OF UNSUITABLE MATERIAL AS SHOWN ON PLANS.
5. SEEDING, FERTILIZING & MULCHING WHEN REQUIRED WILL BE PAID UNDER SECTION 981.



CLEARING LIMITS - TREES AND LOGS

TYPICAL ID	CLEARING METHOD	CLEARING LIMITS			* LEAVE TREES		CLEAR ZONE			STUMPS	HAZARD TREE	DISPOSAL METHOD	COMMENTS
		DOWNHILL WIDTH Ⓐ	UPHILL WIDTH Ⓑ	CLEARING HEIGHT Ⓒ	DISTANCE (FEET) Ⓓ	DIAMETER (INCHES)	DOWNHILL WIDTH Ⓔ	UPHILL WIDTH Ⓕ	MAXIMUM OBSTRUCTION HEIGHT Ⓖ	MAXIMUM HEIGHT Ⓗ	REMOVAL WIDTH Ⓘ		
CLT-1	C	3 FT	3FT	10 FT	2 FT	10 IN	24 IN	24 IN	3 IN	9 IN	45 FT	D 1	



CLEARING METHOD

CLEARING TYPE	CLEARING METHOD	COMMENTS
C1	NEW CONSTRUCTION	TREES, PRUNING, & BRUSH
C2	CLEARING LIMIT RESTORATION	TREES, PRUNING, LOGS, BRUSH & MAINTENANCE
C3	TRAIL OPENING	LOGGING OUT, LOOSE ROCK & DRAINAGE CLEARING
C4	HAZARD TREE REMOVAL	ALONG TRAIL CORRIDOR
C5	HAZARD TREE REMOVAL	INDIVIDUAL (AS MARKED)
C6	LOOSE ROCK & ROOT REMOVAL	X
C7	X	

LEAVE TREES: LEAVE TREES SHOULD BE LIVE, SOUND & UNDAMAGED WITH UNCOMPROMISED ROOT SYSTEMS.

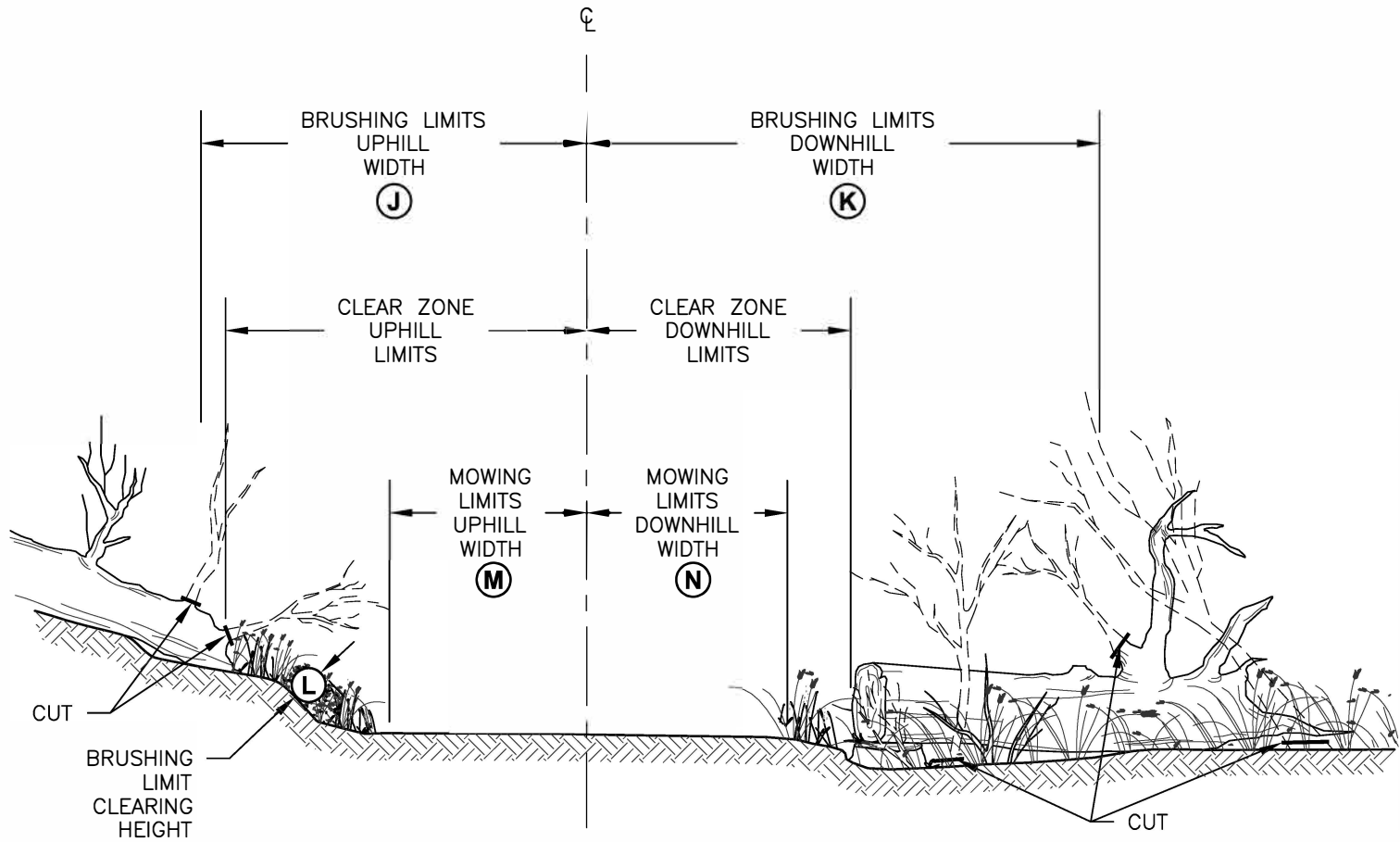
HAZARD TREES: HAZARD TREES ARE TREES THAT ARE STANDING OR LEANING DEAD TREES LARGER THAN 8 INCHES IN DIAMETER AND GREATER THAN 90 FEET IN HEIGHT.

DISPOSAL METHOD

DISPOSAL TYPE	DISPOSAL METHOD	COMMENTS
D1	LOP AND SCATTER OUTSIDE TRAILWAY	X
D2	LOP AND SCATTER ON FILL SLOPE	
D3	PILE AND BURN	
D4	CHIP	
D5	HAUL TO DISPOSAL SITE	
D6		

CLEARING LIMITS - BRUSHING

TYPICAL ID	CLEARING METHOD	BRUSHING LIMITS			MOWING LIMITS		DISPOSAL METHOD	COMMENTS
		UPHILL WIDTH J	DOWNHILL WIDTH K	CLEARING HEIGHT L	UPHILL WIDTH M	DOWNHILL WIDTH N		
CLB-1	C9	3 FT	3 FT	10 FT	-	-	D	



NOTE:
SEE SHEET STD_912-01 FOR
CLEAR ZONE LIMITS

CLEARING METHOD

CLEARING TYPE	CLEARING METHOD	COMMENTS
C9	BRUSHING	
C10	MOWING	
C11		

DISPOSAL METHOD

DISPOSAL TYPE	DISPOSAL METHOD	COMMENTS
D1	LOP AND SCATTER OUTSIDE TRAILWAY	
D2	LOP AND SCATTER ON FILL SLOPE	
D3	PILE AND BURN	
D4	CHIP	
D5	HAUL TO DISPOSAL SITE	
D6		

DRAIN DIP SECTION

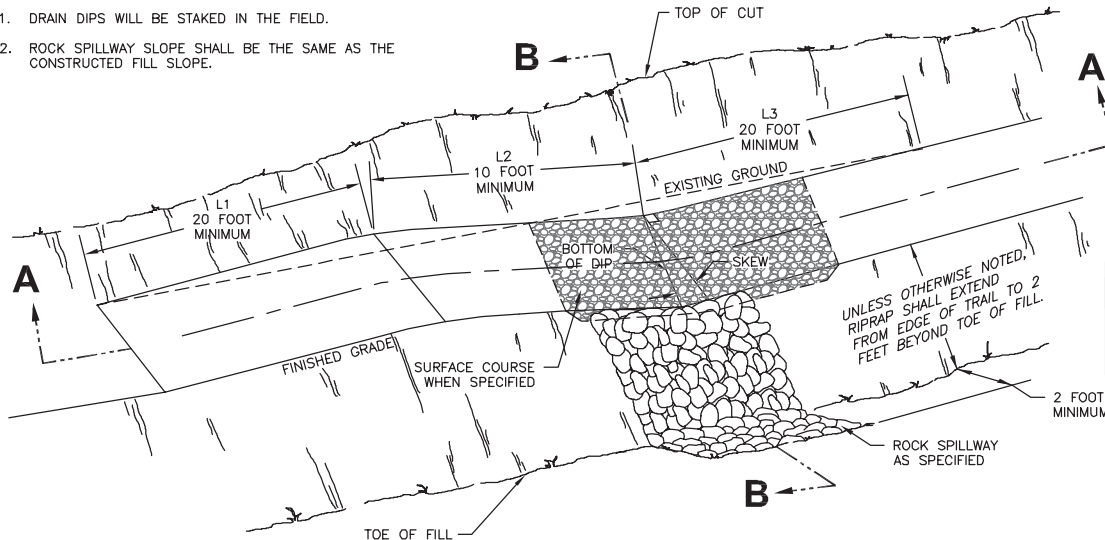
TYPICAL ID	DRAIN DIP TYPE	GEOTEXTILE TYPE	SKEW	SURFACE COURSE		ROCK SPILLWAY***	COMMENTS
				TYPE	DEPTH		
DD1-1	DD1	NA	30 DEG	S 1	6 IN	NA	

N/A WHEN NOT APPLICABLE

***FOR ROCK SPILLWAY SEE SHEET STD_923-10-01

NOTES:

- DRAIN DIPS WILL BE STAKED IN THE FIELD.
- ROCK SPILLWAY SLOPE SHALL BE THE SAME AS THE CONSTRUCTED FILL SLOPE.



GEOTEXTILE TYPE

TYPE	MATERIAL	COMMENTS
G1	NON-WOVEN	
G2	WOVEN	
G3		

SURFACE COURSE MATERIAL TYPE

TYPE	MATERIAL	GRADATION	COMMENTS
S1	PITRUN	1" MINUS	NATIVE MATERIAL
S2	AGGREGATE	X	
S3	X	X	

DRAIN DIP CONSTRUCTION DIMENSIONS

DRAIN DIP TYPE	% PROFILE GRADE	L1	L2	L3	(H)	(E)
DD1	0 TO 4	20 FT	10 FT	20 FT	12 IN	NA
DD2	5 TO 6					
DD3	7 TO 8					
DD4	9 TO 10					
DD5	X					

OVER 10% NOT RECOMMENDED
H = $\frac{1}{2}$ OF TREAD

SHEET 1 OF 2

U.S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE
STANDARD TRAIL PLAN

PROJECT NAME & LOCATION

YOUR PROJECT NAME
YOUR FOREST NAME
YOUR DISTRICT NAME

DRAWING NAME

DRAIN DIP

SECTION

927 - DRAIN DIPS

TYPICAL ID

DD1

REVISION DATE

XX/XX/XX

NO SCALE

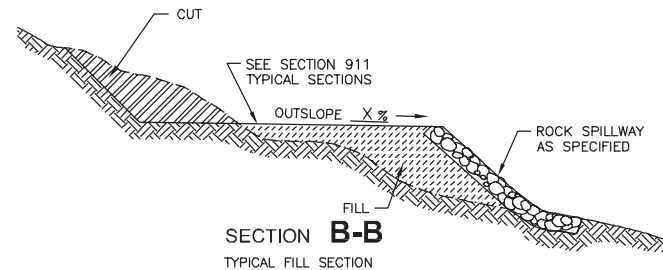
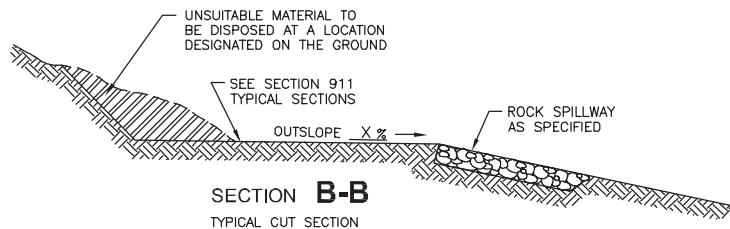
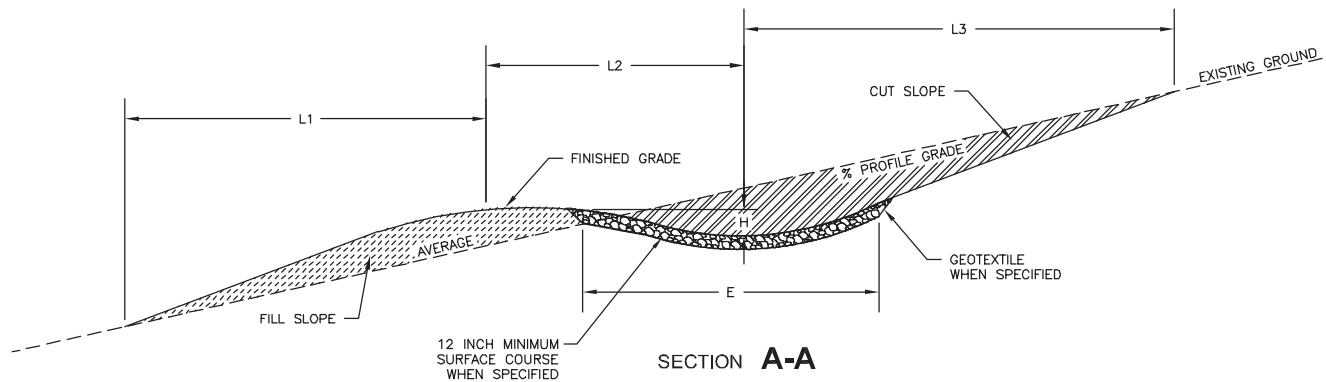
DRAWING NO.

STD_927-10-01

SHEET

16 OF 19

CL
FLOOD LINE



SHEET 2 OF 2

PROJECT NAME & LOCATION

YOUR PROJECT NAME
YOUR FOREST NAME
YOUR DISTRICT NAME

DRAWING NAME

DRAIN DIP DETAILS

SECTION

927 - DRAIN DIPS

TYPICAL ID

DD2

REVISION DATE

XX/XX/XX

NO SCALE

DRAWING NO.

STD_927-10-02

SHEET

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OF

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